

HR Attrition Analysis

A Data-Driven Investigation into Employee Turnover Patterns

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Audience:	Chief Human Resources Officer (CHRO)	Dataset:	750 Employees across 8 Departments

750

EMPLOYEES ANALYSED

8

DEPARTMENTS

115

ATTRITION RECORDS

15.33%

GLOBAL ATTRITION
RATE

01 Business Context

The Chief Human Resources Officer (CHRO) has raised concerns about employee attrition heading into year-end. This portfolio project demonstrates how raw HR data can be translated into targeted business intelligence. By pinpointing **who is leaving, from where, and why**, leadership can direct a finite retention budget toward explicit root causes rather than applying uniform, company-wide interventions.

02 Dataset & Data Quality Assurance

The underlying dataset comprises four relational tables: `employees`, `departments`, `salary_history`, and `attrition`. Before analysis, the data underwent an exhaustive audit to identify structural and input defects.

Quality Issue Detected	Impact / Detail	Resolution Applied in SQL View
Inconsistent Casing	Same roles stored as 'HR Manager', 'HR MANAGER', and 'hr manager '	Applied <code>LOWER(TRIM(job_role))</code> for exact string matching
Orphaned & NULL Dept IDs	~5% of rows mapped to non-existent or blank department keys	Inferred correct <code>department_id</code> using 1-to-1 job role mapping
Duplicate Records	Same person entered twice with disparate <code>employee_id</code> values	Deduplicated via window function <code>ROW_NUMBER()</code> , keeping earliest record
Anomalous Salaries	60 records with negative, zero, or NULL values	Isolated and excluded corrupted fields from the active salary view

DATA STRATEGY OVERVIEW

Two clean, optimized SQL views (`employees_clean` and `salary_clean`) were engineered to resolve all data defects at query runtime. This preserves the original source data while maintaining reproducibility.

03 Analysis & Findings

This section addresses the five business questions using a side-by-side framework: executive strategic takeaways and the underlying technical execution.

Q1: Which departments have the highest attrition rate?

Department	Total Employees	Leavers	Attrition Rate
Sales	85	18	21.18%
Human Resources	98	17	17.35%
Operations	86	14	16.28%
Engineering	88	14	15.91%
Legal	101	16	15.84%
Finance	97	13	13.40%
Customer Support	98	13	13.27%
Marketing	97	9	9.28%

EXECUTIVE TAKEAWAY

Sales has the highest turnover rate at 21.18% (approx. 1 in 5 employees). Marketing is the most stable at 9.28%. The variance between departments confirms that aggregate corporate metrics obscure critical retention localized risks.

TECHNICAL EXECUTION

Aggregated data using a `LEFT JOIN` from `employees_clean` to `attrition`. This preserves active headcount in the denominator, avoiding artificial inflation of the attrition rate calculation.

Q2: Is attrition voluntary or involuntary, and does the split vary?

Department	Exit Type	Leavers	Rate (vs Total Headcount)
Human Resources	Voluntary	14	14.29%
Sales	Voluntary	11	12.94%
Engineering	Voluntary	11	12.50%
Legal	Voluntary	11	10.89%
Operations	Voluntary	9	10.47%
Sales	Involuntary	7	8.24%

EXECUTIVE TAKEAWAY

Human Resources has the highest voluntary turnover (14.29%), indicating internal friction. Sales is highly volatile across both metrics: 12.94% voluntary exits and 8.24% involuntary exits, pointing to aggressive performance pressures.

TECHNICAL EXECUTION

Constructed a Common Table Expression (CTE) to cache absolute department headcount before partition. Grouping blindly by `exit_type` would alter the denominator, skewing the rate metrics.

Q3: Are high performers leaving at a higher rate than low performers?

Performance Rating	Total Employees	Leavers	Attrition Rate
5 — Exceptional	97	19	19.59%
4 — Strong	215	33	15.35%
2 — Below Avg	74	11	14.86%
3 — Average	297	43	14.48%
1 — Poor	37	5	13.51%

CRITICAL ORGANIZATIONAL RISK

Top performers (Rating 5) exhibit the highest attrition rate (19.59%), while the lowest performers are the most stable (13.51%). The company is disproportionately losing its best talent, severely degrading its overall baseline capability over time.

TECHNICAL EXECUTION

Grouped calculations by `performance_rating`. Filtered out `NULL` values (~4% of dataset) using a clear `WHERE performance_rating IS NOT NULL` predicate to avoid compromising the baseline evaluation metrics.

Q4: What is the salary gap between leavers and stayers per job role?

Job Role	Avg Salary (Stayers)	Avg Salary (Leavers)	Variance (Stayers - Leavers)
Compliance Officer	\$705,504	\$867,500	-\$161,996
Operations Manager	\$861,844	\$1,023,567	-\$161,723
DevOps Engineer	\$852,979	\$990,567	-\$137,588
Sales Manager	\$954,396	\$1,054,933	-\$100,537
HR Manager	\$842,941	\$741,100	+\$101,841
Marketing Manager	\$873,638	\$759,580	+\$114,058

EXECUTIVE TAKEAWAY

Most critical roles display a negative salary gap; leavers were earning significantly more than active stayers. This completely rules out base salary as the driver for turnover in Sales and Operations, pointing instead to systemic non-monetary elements (burnout, growth caps).

TECHNICAL EXECUTION

Isolated current pay using window functions ordering by `effective_date DESC`. Calculated conditional averages within a single execution block via `AVG(CASE WHEN...)` to cleanly pivot the comparison matrix.

Q5: Which departments have the longest average tenure before employees leave?

Department	Average Lifespan / Tenure Before Exit
Finance	2.37 years
Legal	2.66 years
Sales	2.80 years
Human Resources	3.42 years
Engineering	3.91 years

EXECUTIVE TAKEAWAY

Finance and Legal employees exit earliest, averaging only 2.37 to 2.66 years. Combined with the compensation profiles, this patterns suggests early onboarding disillusionment or rigid internal hierarchies, rather than active salary disputes.

TECHNICAL EXECUTION

Computed via `DATEDIFF(exit_date, hire_date) / 365.25` . Enforced a strict `WHERE exit_date IS NOT NULL` constraint to guarantee historical calculations reflect actual terminated cohorts only.

04 Strategic Recommendations

Based on the quantitative data, the following tactical actions are recommended, tiered by organizational urgency and financial impact:

URGENT 1. Investigate Sales Management and Quota Structures

Sales presents the highest total attrition (21.18%), matched with highly compensated leavers. This isolates the driver as a culture or target-setting issue, not a salary deficit. The CHRO should conduct immediate structured skip-level reviews and evaluate the equity of performance quotas.

URGENT 2. Address High-Performer Retention Deficits

Exceptional personnel (Rating 5) are leaving at a peak pace of 19.59%. To protect institutional capability, the HR team must audit high-performer career mapping, evaluate total reward packages against external benchmarks, and implement immediate stay-interviews.

MONITOR 3. Remediate Human Resources Voluntary Turnover

The HR department exhibits a high voluntary exit rate of 14.29%. A team designed to retain corporate talent must first be stable itself. Deploy an external, completely anonymous pulse survey tool to safely document and remedy internal leadership issues.

MONITOR 4. Implement Structured Mid-Tenure Reviews in Finance and Legal

With an average tenure under 2.7 years, corporate professionals are leaving before optimizing their operational value. Introduce formalized 18-month check-ins and clearer promotional tracks to intercept early career fatigue.

MAINTAIN 5. Document and Export Marketing Department Best Practices

Marketing has maintained an exceptional attrition floor of 9.28%. The CHRO should systematically audit their internal management style, operational workflows, and team culture to establish a standard framework for struggling units.

05 Technical Methodology & Analytical Limitations

To preserve analytical transparency and project documentation, the architectural constraints are outlined below:

- **Tool Integration:** Engineered entirely within MySQL (v8.0) for core data manipulation, view scheduling, and primary window calculations. Document layout compiled via Python/WeasyPrint HTML-to-PDF pipeline.
- **Granularity Constraints:** Attrition rates are computed using a static snapshot total denominator rather than a rolling 12-month average, due to limitations in historical daily headcount logs.
- **Irrecoverable Gaps:** Three terminated employee files (~2.6% of exits) lacked accompanying metadata in the source `attrition` table and were excluded from exit-type partition to safeguard structural integrity.
- **Compensation Scope:** Calculations are restricted exclusively to base salary logs and do not incorporate variable parameters (bonuses, equity, commissions), which represent heavy retention metrics in Sales roles.